

EMMEN AP, Switzerland

WMO: 066130

Lat: 47.084N

Lon: 8.296E

Elev: 1400

StdP: 13.97

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.4	19.1	10.6	9.9	16.1	15.5	12.7	21.5	18.6	46.3	16.1	38.9	4.3	30	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	86.4	69.2	83.0	67.7	79.3	66.0	70.5	83.5	68.6	80.1	67.2	77.6	5.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	101.1	77.2	64.4	95.5	74.9	62.8	90.2	72.7	35.3	83.7	33.7	80.7	32.5	77.8	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	11.9	10.1	7.4	91.7	5.1	3.2	3.7	93.9	0.7	95.8	-2.2	97.6	-6.0	99.9		
			7.7	73.7	4.6	2.5	4.4	75.5	1.7	77.0	-0.9	78.4	-4.2	80.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	33.4	33.7	41.1	48.8	55.7	63.4	66.4	65.0	58.3	50.7	40.6	34.9	(6)
	DBStd	13.56	7.58	7.32	6.56	6.84	6.83	6.03	5.70	5.53	5.47	6.26	6.68	6.31	(7)
	HDD50	2210	515	456	280	102	26	1	0	0	4	71	288	467	(8)
	HDD65	5915	979	876	742	485	296	101	51	68	208	444	733	932	(9)
	CDD50	2002	2	0	3	67	202	402	509	466	254	92	5	0	(10)
	CDD65	232	0	0	0	0	7	53	95	69	8	0	0	0	(11)
	CDH74	2395	0	0	0	19	121	567	962	649	75	2	0	0	(12)
	CDH80	679	0	0	0	0	22	158	319	172	8	0	0	0	(13)
(14) Wind	WSAvg	3.2	3.1	3.1	3.5	3.6	3.8	3.6	3.5	2.9	2.7	2.6	2.6	3.0	(14)
(15) Precipitation	PrecAvg	59.4	3.2	3.5	3.7	4.6	6.5	6.9	6.9	6.9	4.9	4.1	4.2	4.1	(15)
	PrecMax	73.5	7.1	8.5	9.4	9.3	11.3	10.2	13.4	13.6	9.4	7.1	9.8	9.1	(16)
	PrecMin	46.2	0.4	1.4	1.3	0.8	1.5	3.7	2.8	1.7	1.8	0.4	0.0	0.1	(17)
	PrecStd	6.8	1.8	1.8	1.9	2.1	2.6	1.7	2.7	2.7	1.9	1.6	2.1	2.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	57.1	66.5	76.8	82.5	89.7	91.3	89.6	80.6	71.9	62.5	53.4	(19)
		MCWB	51.3	45.7	51.7	57.4	63.5	70.6	69.4	71.3	67.5	60.3	53.0	46.9	(20)
	2%	DB	52.0	52.1	62.4	73.0	77.1	84.3	87.7	84.6	75.6	67.6	57.6	49.8	(21)
		MCWB	46.0	43.5	50.2	56.8	61.2	68.3	69.1	69.7	64.3	58.6	51.3	44.7	(22)
	5%	DB	47.8	48.2	58.6	68.3	73.4	80.6	84.0	80.7	73.0	64.3	53.7	46.7	(23)
		MCWB	43.3	41.9	48.3	54.4	59.7	66.9	67.8	68.2	63.0	57.6	48.9	43.2	(24)
	10%	DB	43.4	44.4	53.9	64.2	69.6	76.7	79.2	77.1	69.6	60.9	50.3	44.4	(25)
		MCWB	40.7	40.2	46.1	52.5	58.0	64.5	65.7	66.0	61.3	55.9	46.9	41.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	47.1	52.9	59.6	65.7	72.7	72.6	72.8	67.6	62.0	54.7	48.7	(27)
		MCDB	57.9	54.1	63.9	73.1	80.3	86.6	87.5	86.8	78.1	68.7	60.7	52.0	(28)
	2%	WB	46.9	44.0	50.7	57.3	63.0	69.6	70.2	70.4	65.3	59.9	51.9	46.0	(29)
		MCDB	51.5	49.8	60.4	70.7	74.1	81.5	83.9	82.7	74.0	65.1	56.7	48.6	(30)
	5%	WB	43.5	42.1	48.8	55.3	60.8	67.4	68.2	68.3	63.6	58.1	49.6	43.6	(31)
		MCDB	46.4	47.4	57.4	67.3	70.2	78.2	80.6	78.7	71.3	62.9	53.1	45.9	(32)
	10%	WB	40.7	39.9	46.7	53.1	58.7	65.3	66.8	66.9	61.9	56.4	47.6	41.5	(33)
		MCDB	43.1	43.8	53.6	62.5	68.0	75.1	77.9	75.9	68.4	60.5	49.9	43.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	14.2	19.0	21.4	19.1	20.0	20.1	19.4	18.2	16.5	13.0	13.7	(35)
		MCDBR	16.5	22.5	28.1	30.6	27.5	26.9	28.4	25.8	23.5	21.7	19.6	16.9	(36)
	5% WB	MCWBR	11.4	14.8	17.1	16.8	13.6	13.0	12.2	12.8	13.3	13.6	13.4	12.1	(37)
		MCWBR	15.5	20.4	25.9	29.1	24.1	25.1	25.5	24.6	21.7	18.9	17.5	15.4	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.294	0.324	0.370	0.410	0.427	0.432	0.434	0.422	0.392	0.364	0.324	0.291	(40)
		taud	2.369	2.289	2.231	2.183	2.194	2.234	2.228	2.273	2.335	2.386	2.425	2.391	(41)
	Ebn at Noon	Ebn at Noon	251	265	267	267	266	265	263	261	258	247	238	238	(42)
		Edh at Noon	23	30	37	43	44	43	43	39	34	28	22	20	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	415	694	1077	1389	1563	1820	1777	1547	1180	757	435	342	(44)
	RadStd	RadStd	49	91	152	209	155	167	159	149	117	82	57	51	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (120 neighbors)	N/A	N/A	N/A	+1.28	+0.74	+0.58	N/A	N/A	+103	+27

Nomenclature: See separate page